Sulabham · APPSC AEE 2026 — Module 2 Test: Fluid Mechanics and Hydraulic Machines

Paper 2 (Civil & Mechanical common) · original practice questions in the pattern of APPSC AEE Paper II, 22 Aug 2023 · take at the end of Week 6

Time: 50 minutes (exam pace: 1 minute per question) **Marks:** +1 correct, −⅓ wrong, 0 unattempted
Calculator: not allowed
Write your answers on a separate sheet. Mark a question only if you can rule out at least two options.
Online version with timer: sulabham.in/AEE_Module2_Test.html

Questions

A. Properties of Fluids

1. The excess pressure inside a soap bubble of 40 mm diameter, if the surface tension of the soap solution is 0.04 N/m, is:
(a) 2 N/m² (b) 4 N/m² (c) 8 N/m² (d) 16 N/m²

2. Statement 1: Kinematic viscosity is the ratio of dynamic viscosity to mass density.
Statement 2: The CGS unit of kinematic viscosity is the stoke, equal to 1 cm²/s.
(a) Both statements are true (b) Both statements are false
(c) Statement 1 is true, Statement 2 is false (d) Statement 1 is false, Statement 2 is true

B. Pressure and Its Measurement

3. The gauge pressure at a depth of 5 m below the free surface of an oil of specific gravity 0.8 is (take ρg of water = 9810 N/m³):
(a) 31.39 kN/m² (b) 39.24 kN/m² (c) 49.05 kN/m² (d) 61.31 kN/m²

4. In a hydraulic press, the ram is 200 mm in diameter and the plunger is 20 mm in diameter. A force of 100 N applied on the plunger lifts a load of:
(a) 100 kN (b) 1 kN (c) 2 kN (d) 10 kN

5. A simple U-tube mercury manometer is connected to a pipe carrying water. The centre of the pipe is 0.2 m above the mercury level in the left limb, and the mercury in the right limb (open to atmosphere) stands 0.3 m above that level. The gauge pressure head at the centre of the pipe is:
(a) 4.08 m of water (b) 4.28 m of water (c) 3.88 m of water (d) 0.50 m of water

6. Which of the following statements is INCORRECT?
(a) A piezometer can be used to measure vacuum (negative gauge) pressure
(b) Absolute pressure = atmospheric pressure + gauge pressure
(c) Standard atmospheric pressure is 760 mm of mercury
(d) Mechanical gauges such as the Bourdon tube are used for high pressures

C. Hydrostatic Forces, Buoyancy and Floating Bodies

7. A circular plate of 2 m diameter is immersed vertically in water with its centre 3 m below the free surface. The total pressure force and the depth of the centre of pressure are:
(a) 92.5 kN and 3.00 m (b) 92.5 kN and 3.083 m (c) 61.6 kN and 3.083 m (d) 92.5 kN and 3.25 m

8. A quadrant gate of radius 2 m and width 1 m holds back water whose free surface is level with the top of the gate. The horizontal and vertical components of the hydrostatic force on the gate are:
(a) 30.82 kN and 19.62 kN (b) 19.62 kN and 19.62 kN (c) 39.24 kN and 30.82 kN (d) 19.62 kN and 30.82 kN

9. A body weighs 300 N in air and 200 N when fully immersed in water. Its specific gravity is:
(a) 1.5 (b) 2.0 (c) 3.0 (d) 0.67

10. A rectangular pontoon 5 m long and 3 m wide floats in water with a depth of immersion of 0.8 m. Its centre of gravity is 0.6 m above the base. The metacentric height is about:
(a) 0.54 m (b) 0.94 m (c) 1.14 m (d) 0.74 m

11. Consider the following statements:

I. A floating body is in stable equilibrium when its metacentre lies above its centre of gravity.

II. A fully submerged body is in stable equilibrium when its centre of buoyancy lies above its centre of gravity.

III. Raising the centre of gravity of a floating body increases its metacentric height.

IV. A larger metacentric height gives a longer period of rolling.

Which of the above statements are correct?

(a) I and II only (b) I, II and III only (c) II and IV only (d) I, II and IV only

D. Fluid Kinematics

12. A pipe of 300 mm diameter carrying water at 2 m/s branches into two pipes of 150 mm and 200 mm diameter. If the velocity in the 150 mm pipe is 3 m/s, the velocity in the 200 mm pipe is about:

(a) 2.00 m/s (b) 2.81 m/s (c) 3.50 m/s (d) 4.50 m/s

13. For a steady, two-dimensional, incompressible flow, $u = 4xy$. If $v = 0$ at $y = 0$, the velocity component v is:

(a) $-4y^2$ (b) $2y^2$ (c) $-2y^2$ (d) $-2x^2$

14. The stream function of a two-dimensional flow is $\psi = 2xy$. The magnitude of the velocity at the point (1, 2) is:

(a) 2.00 units (b) 3.46 units (c) 6.00 units (d) 4.47 units

15. For the two-dimensional flow with stream function $\psi = x^2 - y^2$, consider the following statements:

I. The flow is irrotational.

II. The magnitude of velocity at the point (1, 1) is $2\sqrt{2}$ units.

III. A velocity potential function exists for this flow.

Which of the above statements are correct?

(a) I only (b) I and III only (c) II and III only (d) I, II and III

16. For the flow field $u = 2y$, $v = 3x$, the vorticity is:

(a) 1.0 (b) 0.5 (c) 5.0 (d) 2.5

17. Water flows at a constant discharge through a pipe whose diameter reduces gradually along its length. The flow is:

(a) steady and uniform (b) steady and non-uniform (c) unsteady and uniform (d) unsteady and non-uniform

18. In a steady two-dimensional flow, $u = 2x$ and $v = -2y$. The magnitude of the acceleration of a fluid particle at the point (1, 1) is:

(a) 4.00 units (b) 8.00 units (c) 5.66 units (d) 2.83 units

19. Assertion (A): In steady flow, streamlines, pathlines and streaklines coincide.

Reason (R): In steady flow, the velocity at any given point does not change with time.

(a) Both A and R are true, and R is the correct explanation of A (b) Both A and R are true, but R is NOT the correct explanation of A

(c) A is true, but R is false (d) A is false, but R is true

20. Which of the following statements about a flow net is INCORRECT?

(a) Streamlines and equipotential lines intersect each other at right angles (b) A flow net can be drawn for a rotational flow

(c) It is used to estimate seepage and uplift pressure under hydraulic structures (d) Where the streamlines are closer together, the velocity is higher

E. Fluid Dynamics and Flow Measurement

21. Water flows through a horizontal pipe that tapers from 200 mm to 100 mm diameter. At the 200 mm section the pressure is 100 kN/m² and the velocity is 2 m/s. Neglecting losses, the pressure at the 100 mm section is:

(a) 130 kN/m² (b) 94 kN/m² (c) 40 kN/m² (d) 70 kN/m²

22. A venturimeter has an inlet-to-throat area ratio of 4. The difference of piezometric head between the inlet and the throat is 0.75 m of the flowing liquid. Taking $g = 10$ m/s², the theoretical velocity at the throat is:

(a) 3.87 m/s (b) 4.00 m/s (c) 4.24 m/s (d) 16.0 m/s

23. A differential mercury manometer connected to a venturimeter carrying oil of specific gravity 0.8 shows a reading of 20 cm. The difference of pressure head is:

(a) 2.52 m of oil (b) 2.72 m of oil (c) 3.20 m of oil (d) 3.40 m of oil

24. Consider the following statements:

I. The coefficient of discharge of a venturimeter is higher than that of an orifice meter.

II. An orifice meter is cheaper and needs less space than a venturimeter.

III. The loss of head in an orifice meter is smaller than in a venturimeter.

IV. Both meters work on Bernoulli's principle.

Which of the above statements are correct?

(a) I and II only (b) I, II and IV only (c) II, III and IV only (d) I, II, III and IV

25. A pitot tube placed in a pipe shows a stagnation head of 6 m and a static head of 5 m of water. Taking the coefficient of the tube as 1 and $g = 9.81\ m/s^2$, the velocity at that point is:

(a) 4.43 m/s (b) 3.13 m/s (c) 9.81 m/s (d) 19.62 m/s

26. Match List-I with List-II:

List-I	**List-II**
A. Kinetic energy correction factor, laminar pipe flow	1. About 0.62
B. Momentum correction factor, laminar pipe flow	2. 4/3
C. Coefficient of discharge of a venturimeter	3. About 0.98
D. Coefficient of discharge of an orifice meter	4. 2

(a) A-2, B-4, C-3, D-1 (b) A-4, B-2, C-1, D-3 (c) A-4, B-2, C-3, D-1 (d) A-2, B-4, C-1, D-3

27. In a long horizontal pipe of uniform diameter carrying a steady flow, the vertical distance between the total energy line and the hydraulic gradient line:

(a) remains constant along the pipe (b) increases in the direction of flow (c) decreases in the direction of flow (d) is zero

28. Statement 1: Euler's equation of motion is obtained by applying Newton's second law to a fluid element, considering only pressure and gravity forces.

Statement 2: Bernoulli's equation is obtained by integrating Euler's equation along a streamline for an incompressible fluid.

(a) Both statements are true (b) Both statements are false

(c) Statement 1 is true, Statement 2 is false (d) Statement 1 is false, Statement 2 is true

F. Flow Through Pipes

29. Water flows at 2 m/s through a pipe 1000 m long and 200 mm in diameter. If the Darcy friction factor f is 0.02 ($h_f = fLV^2/2gD$) and $g = 10\ m/s^2$, the loss of head due to friction is:

(a) 5 m (b) 10 m (c) 20 m (d) 80 m

30. For laminar flow through a circular pipe at a Reynolds number of 1600, the Darcy friction factor is:

(a) 0.01 (b) 0.004 (c) 0.4 (d) 0.04

31. Two pipes of the same length and the same friction factor, of diameters D and 2D, are connected in parallel between two reservoirs. The ratio of the discharge in the larger pipe to that in the smaller pipe is:

(a) 5.66 (b) 2 (c) 4 (d) 32

32. Water flows from a larger pipe into a smaller pipe in which the velocity is 4 m/s. If the coefficient of contraction is 0.625 and $g = 10\ m/s^2$, the loss of head due to the sudden contraction is:

(a) 0.400 m (b) 0.288 m (c) 0.800 m (d) 0.480 m

33. A pipe supplies water to a nozzle from a reservoir under a total head of 240 m. For maximum transmission of power, the loss of head due to friction and the efficiency of transmission are:

(a) 120 m and 50 % (b) 80 m and 33.3 % (c) 60 m and 75 % (d) 80 m and 66.7 %

34. Water flows at 1.5 m/s in a rigid pipe 2100 m long. The valve at the end is closed completely in 2 s. If the velocity of the pressure wave is 1400 m/s, the rise in pressure is:

(a) 157.5 N/cm^2 (b) 105 N/cm^2 (c) 210 N/cm^2 (d) 420 N/cm^2

35. Assertion (A): In turbulent flow through a pipe, the loss of head is approximately proportional to the square of the velocity.
Reason (R): In turbulent flow, the Darcy friction factor is equal to 64/Re.
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is NOT the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

G. Boundary Layer, Drag and Lift

36. Air ($\nu = 1.5 \times 10^{-5}$ m²/s) flows at 1.5 m/s over a thin flat plate. Using $\delta = 5x/\sqrt{Re_x}$, the thickness of the laminar boundary layer at 0.4 m from the leading edge is:
(a) 10 mm (b) 2 mm (c) 5 mm (d) 20 mm

37. The velocity profile in a boundary layer of thickness 12 mm is linear, $u/U = y/\delta$. The displacement thickness and the momentum thickness are:
(a) 4 mm and 2 mm (b) 6 mm and 2 mm (c) 6 mm and 3 mm (d) 2 mm and 6 mm

38. Match List-I with List-II (limits of integration 0 to δ):

List-I	List-II
A. Displacement thickness	1. $\int (u/U)(1 - u^2/U^2)\, dy$
B. Momentum thickness	2. Displacement thickness ÷ momentum thickness
C. Energy thickness	3. $\int (1 - u/U)\, dy$
D. Shape factor	4. $\int (u/U)(1 - u/U)\, dy$

(a) A-4, B-3, C-1, D-2 (b) A-3, B-1, C-4, D-2 (c) A-3, B-4, C-2, D-1 (d) A-3, B-4, C-1, D-2

39. Consider the following statements:
I. For a well-streamlined body, the pressure drag is small compared with the friction drag.
II. For a bluff body, the pressure (form) drag is the major part of the total drag.
III. The drag force acts perpendicular to the direction of the free stream.
IV. Stokes' law for the drag on a sphere is valid only at very low Reynolds numbers.
Which of the above statements are correct?
(a) I and II only (b) I, II and III only (c) I, II and IV only (d) I, II, III and IV

40. A kite of area 2 m² is held in a wind of 10 m/s. If the coefficient of lift is 0.8 and the density of air is 1.2 kg/m³, the lift force is:
(a) 9.6 N (b) 48 N (c) 192 N (d) 96 N

H. Impact of Jets and Hydraulic Turbines

41. A jet of water of cross-sectional area 0.002 m² and velocity 10 m/s strikes a stationary flat plate inclined at 30° to the axis of the jet. The force normal to the plate is:
(a) 50 N (b) 100 N (c) 173 N (d) 200 N

42. A Pelton wheel works under a net head of 180 m. Taking the coefficient of velocity of the nozzle as 1 and g = 10 m/s², the theoretical bucket speed for maximum efficiency is:
(a) 15 m/s (b) 42.4 m/s (c) 60 m/s (d) 30 m/s

43. A turbine runs at 200 rpm and discharges 9 m³/s under a head of 25 m. If the head falls to 16 m, with the same efficiency its speed and discharge become:
(a) 160 rpm and 7.2 m³/s (b) 128 rpm and 5.76 m³/s (c) 160 rpm and 5.76 m³/s (d) 250 rpm and 11.25 m³/s

44. Which of the following statements is INCORRECT?
(a) A Pelton wheel is a tangential-flow impulse turbine
(b) A modern Francis turbine is an inward mixed-flow reaction turbine
(c) The runner blades of a Kaplan turbine are fixed and cannot be adjusted
(d) A draft tube is used with reaction turbines

45. Statement 1: The specific speed of a turbine is given by N√P / H^(5/4).
Statement 2: Among Pelton, Francis and Kaplan turbines, the Pelton wheel has the highest specific speed.
(a) Both statements are true
(b) Both statements are false
(c) Statement 1 is true, Statement 2 is false
(d) Statement 1 is false, Statement 2 is true

I. Centrifugal Pumps and Control Valves

46. A centrifugal pump runs at 1000 rpm and delivers 0.16 m^3/s against a manometric head of 16 m. Its specific speed ($N\sqrt{Q} / H^{(3/4)}$) is:

(a) 25 (b) 50 (c) 100 (d) 400

47. A centrifugal pump develops a manometric head of 18 m. The whirl velocity at the outlet is 12 m/s and the tangential velocity of the impeller at the outlet is 20 m/s. Taking g = 10 m/s^2, the manometric efficiency is:

(a) 60 % (b) 133 % (c) 90 % (d) 75 %

48. A pump set has to deliver water against a total head of 150 m. Each identical impeller develops 37.5 m at the design discharge. The arrangement needed is:

(a) Four impellers in parallel (b) Four impellers in series (c) Two impellers in series (d) Two impellers in parallel

49. Assertion (A): A foot valve is fitted at the lower end of the suction pipe of a centrifugal pump.
Reason (R): It is a non-return valve that keeps the suction pipe and casing full of water when the pump is stopped.

(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is NOT the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

J. Hydro-Power Projects

50. Consider the following statements:

I. A surge tank protects the water conductor system against water hammer.
II. Pondage means the seasonal storage of monsoon flows in a large reservoir.
III. A mass curve is used to find the storage capacity of a reservoir.
IV. A spillway disposes of surplus flood water safely.

Which of the above statements are correct?

(a) I and II only (b) I, III and IV only (c) II, III and IV only (d) I, II, III and IV

Answer key with short solutions

A. Properties of Fluids

1. (c) Soap bubble has two surfaces: $p = 8\sigma/d = 8 \times 0.04/0.04 = 8$ N/m². (4 N/m² is the droplet formula $4\sigma/d$ — the trap.)

2. (a) $\nu = \mu/\rho$, and 1 stoke = 1 cm²/s = 10^{-4} m²/s. Both true. (Poise is the CGS unit of dynamic viscosity.)

B. Pressure and Its Measurement

3. (b) $p = S \times 9810 \times h = 0.8 \times 9810 \times 5 = 39{,}240$ N/m² = 39.24 kN/m². (49.05 is for water; 61.31 divides by 0.8 instead of multiplying.)

4. (d) Pascal's law: $W = F \times (D/d)^2 = 100 \times (200/20)^2 = 100 \times 100 = 10{,}000$ N = 10 kN. (1 kN uses the diameter ratio instead of the area ratio.)

5. (c) Equate at the lower mercury level: $h_A + 0.2 = 13.6 \times 0.3 \rightarrow h_A = 4.08 - 0.2 = 3.88$ m of water. (4.08 forgets the water column; 4.28 adds it instead of subtracting.)

6. (a) Under vacuum, air would be sucked into an open piezometer tube; it only reads positive gauge pressure of liquids. The other three are correct.

C. Hydrostatic Forces, Buoyancy and Floating Bodies

7. (b) $F = \rho g A\bar{x} = 9810 \times \pi \times 1^2 \times 3 = 92.5$ kN. $h^* = \bar{x} + I_G/(A\bar{x}) = 3 + (d^2/16)/3 = 3 + 0.25/3 = 3.083$ m. The centre of pressure is always below the centroid.

8. (d) F_H = force on the vertical projection (2 m × 1 m, $\bar{x} = 1$ m) = $9810 \times 2 \times 1 = 19.62$ kN. F_V = weight of water in the quadrant = $9810 \times (\pi \times 2^2/4) \times 1 = 30.82$ kN.

9. (c) Loss of weight = buoyant force = 100 N = weight of an equal volume of water. S = weight in air / loss of weight = 300/100 = 3.

10. (d) $BM = I/V = (5 \times 3^3/12)/(5 \times 3 \times 0.8) = 11.25/12 = 0.94$ m. B is at 0.4 m above the base, so $BG = 0.6 - 0.4 = 0.2$ m. $GM = BM - BG = 0.74$ m. (0.94 is BM alone; 1.14 adds BG.)

11. (a) III is false: raising G increases BG, so GM = BM − BG falls. IV is false: $T = 2\pi\sqrt{(k^2/(g \cdot GM))}$, so a larger GM gives a shorter, jerkier roll.

D. Fluid Kinematics

12. (b) Continuity: $d_1^2V_1 = d_2^2V_2 + d_3^2V_3 \rightarrow 0.09 \times 2 = 0.0225 \times 3 + 0.04 \times V_3 \rightarrow V_3 = (0.18 - 0.0675)/0.04 = 2.81$ m/s.

13. (c) Continuity: $\partial u/\partial x + \partial v/\partial y = 0 \rightarrow 4y + \partial v/\partial y = 0 \rightarrow v = -2y^2 + f(x)$; v = 0 at y = 0 gives f(x) = 0. (2023 Q115 asked for a missing component in the same way.)

14. (d) $u = -\partial\psi/\partial y = -2x = -2$; $v = \partial\psi/\partial x = 2y = 4$. $|V| = \sqrt{(4 + 16)} = 4.47$ units. The magnitude does not depend on the sign convention used.

15. (d) $\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = 2 - 2 = 0 \rightarrow$ irrotational, so φ exists. $u = -\partial\psi/\partial y = 2y = 2$, $v = \partial\psi/\partial x = 2x = 2 \rightarrow |V| = 2\sqrt{2}$. All three are correct.

16. (a) Vorticity = $\partial v/\partial x - \partial u/\partial y = 3 - 2 = 1$. Rotation $\omega = \frac{1}{2} \times$ vorticity = 0.5 (the trap). 2.5 is the shear strain rate $\frac{1}{2}(3 + 2)$.

17. (b) Discharge constant with time → steady. Velocity changes from section to section ($V \propto 1/d^2$) → non-uniform. (2023 Q107 asked the draining-tank case: unsteady and uniform.)

18. (c) $a_x = u\,\partial u/\partial x + v\,\partial u/\partial y = 2x \times 2 = 4$; $a_y = u\,\partial v/\partial x + v\,\partial v/\partial y = (-2y)(-2) = 4$. $|a| = 4\sqrt{2} = 5.66$. The flow is steady, yet the convective acceleration is not zero.

19. (a) Because the velocity field is fixed in time, every particle passing a point follows the same streamline, so all three lines are identical. R explains A.

20. (b) Equipotential lines exist only when a velocity potential exists, i.e. for irrotational flow. A flow net cannot be drawn for rotational flow or where the flow separates (2023 Q104, Q105).

E. Fluid Dynamics and Flow Measurement

21. (d) $V_2 = 2 \times (200/100)^2 = 8$ m/s. $p_2 = p_1 + \rho(V_1^2 - V_2^2)/2 = 100{,}000 + 500 \times (4 - 64) = 70{,}000$ N/m² = 70 kN/m². (94 uses $V_2 = 4$ m/s — the diameter ratio instead of the area ratio.)

22. (b) $V_2 = \sqrt{[2gh/(1 - (a_2/a_1)^2)]} = \sqrt{[15/(1 - 1/16)]} = \sqrt{16} = 4$ m/s. (3.87 = $\sqrt{(2gh)}$ ignores the velocity of approach.)

23. (c) h = x(S_Hg/S_oil − 1) = 0.2 × (13.6/0.8 − 1) = 0.2 × 16 = 3.2 m of oil. (2.52 uses the factor 12.6, which is only for water.)

24. (b) III is false: the sudden contraction and expansion at an orifice plate cause a larger permanent head loss; the venturimeter's gradual cones keep the loss small ($C_d \approx 0.98$ against ≈ 0.62).

25. (a) $V = \sqrt{(2g\,\Delta h)} = \sqrt{(2 \times 9.81 \times 1)} = 4.43$ m/s. A pitot tube gives the velocity at a point, not the discharge.

26. (c) Laminar flow in a pipe: $\alpha = 2$, $\beta = 4/3$ (for turbulent flow both are close to 1). Venturimeter $C_d \approx 0.95$–0.99; orifice meter $C_d \approx 0.6$–0.65.

27. (a) The gap between TEL and HGL is the velocity head $V^2/2g$. With a constant diameter and discharge, V is constant, so the two lines are parallel and both slope down due to friction.

28. (a) Euler's equation: $dp/\rho + g\,dz + V\,dV = 0$ (viscous forces neglected). Integrating with ρ constant gives $p/\rho g + V^2/2g + z =$ constant. Both true.

F. Flow Through Pipes

29. (c) $h_f = fLV^2/(2gD) = 0.02 \times 1000 \times 4/(2 \times 10 \times 0.2) = 80/4 = 20$ m. (80 m wrongly uses 4f — check which form of the formula the question gives.)

30. (d) $f = 64/Re = 64/1600 = 0.04$. (0.01 is 16/Re, the friction coefficient f' used in the $4f'LV^2/2gD$ form.)

31. (a) Same head loss in parallel: $h_f \propto Q^2/D^5 \rightarrow Q \propto D^{2.5}$. Ratio $= 2^{2.5} = 5.66$. (32 $= 2^5$ forgets the square root; 4 uses the area ratio only.)

32. (b) $h_c = (1/C_c - 1)^2\, V_2^2/2g = (1.6 - 1)^2 \times 16/20 = 0.36 \times 0.8 = 0.288$ m. (0.4 m $= 0.5V^2/2g$ is used only when C_c is not given.)

33. (d) Power is maximum when $h_f = H/3 = 80$ m. Efficiency $= (H - h_f)/H = 160/240 = 66.7$ %. (Asked twice in 2023: Q96 and Q120.)

34. (c) Critical time $2L/C = 2 \times 2100/1400 = 3$ s. Closure in 2 s < 3 s → sudden closure: $p = \rho VC = 1000 \times 1.5 \times 1400 = 2.1 \times 10^6$ N/m² $= 210$ N/cm². (157.5 is the gradual-closure formula $\rho LV/t$.)

35. (c) A is true ($h_f \propto V^{1.75}$ to V^2). R is false: $f = 64/Re$ holds only for laminar flow, where the loss is proportional to V, not V^2.

G. Boundary Layer, Drag and Lift

36. (a) $Re_x = Ux/\nu = 1.5 \times 0.4/(1.5 \times 10^{-5}) = 4 \times 10^4$; $\sqrt{Re_x} = 200$. $\delta = 5 \times 0.4/200 = 0.01$ m = 10 mm. (Re_x is below 5×10^5, so the layer is laminar.)

37. (b) Linear profile: $\delta^* = \delta/2 = 6$ mm, $\theta = \delta/6 = 2$ mm (energy thickness $= \delta/4 = 3$ mm). 2023 Q141 asked $\theta = \delta/6$.

38. (d) $\delta^* = \int(1 - u/U)dy$; $\theta = \int(u/U)(1 - u/U)dy$; energy thickness $= \int(u/U)(1 - u^2/U^2)dy$; shape factor $H = \delta^*/\theta$.

39. (c) III is false: drag acts along the flow direction; lift acts perpendicular to it. Stokes' law $F = 3\pi\mu DV$ holds for Re below about 1 (strictly below 0.2).

40. (d) $F_L = C_L\, A\, \rho U^2/2 = 0.8 \times 2 \times 1.2 \times 100/2 = 96$ N. (192 N forgets the ½.)

H. Impact of Jets and Hydraulic Turbines

41. (b) $F_n = \rho aV^2 \sin\theta = 1000 \times 0.002 \times 100 \times 0.5 = 100$ N. (200 N is for a plate normal to the jet; 173 N uses cos 30°.)

42. (d) Jet velocity $V = \sqrt{(2gH)} = \sqrt{3600} = 60$ m/s. Maximum efficiency when $u = V/2 = 30$ m/s. (In practice the speed ratio is about 0.46.)

43. (a) Unit quantities: $N \propto \sqrt{H}$ and $Q \propto \sqrt{H}$. $\sqrt{(16/25)} = 0.8 \rightarrow N = 160$ rpm, $Q = 7.2$ m³/s. (Power $\propto H^{1.5}$ would fall to 0.512 times.)

44. (c) Kaplan runner blades are adjustable while running (double regulation with the guide vanes). A propeller turbine is the one with fixed blades.

45. (c) Statement 2 is false: Pelton has the lowest specific speed (about 10–60), Francis 60–300, Kaplan 300–1000 (2023 Q121).

I. Centrifugal Pumps and Control Valves

46. (b) $N_s = 1000 \times \sqrt{0.16} / 16^{0.75} = 1000 \times 0.4/8 = 50$. (Pump: $\sqrt{Q}$ and $H^{3/4}$. Turbine: $\sqrt{P}$ and $H^{5/4}$.)

47. (d) $\eta_{mano} = gH_m/(V_{w2}\, u_2) = 10 \times 18/(12 \times 20) = 180/240 = 75$ %. (133 % is the ratio inverted — an efficiency cannot exceed 100 %.)

48. (b) Series (multistage) arrangement adds the heads: 150/37.5 = 4 impellers in series, each passing the same discharge. Parallel adds discharge, not head (2023 Q137).

49. (a) A centrifugal pump cannot lift water unless it is primed. The foot valve (with strainer) holds the priming water, so R correctly explains A.

J. Hydro-Power Projects

50. (b) II is false: pondage is short-term (hours to about a week) storage to meet load fluctuations; long-term seasonal holding is called storage.

Score analysis — fill this in and revise the weakest section first

Section	Questions	Max	Correct	Wrong	Net (C − W/3)
A. Properties of Fluids	1–2	2			
B. Pressure and Its Measurement	3–6	4			
C. Hydrostatic Forces, Buoyancy and Floating Bodies	7–11	5			
D. Fluid Kinematics	12–20	9			
E. Fluid Dynamics and Flow Measurement	21–28	8			
F. Flow Through Pipes	29–35	7			
G. Boundary Layer, Drag and Lift	36–40	5			
H. Impact of Jets and Hydraulic Turbines	41–45	5			
I. Centrifugal Pumps and Control Valves	46–49	4			
J. Hydro-Power Projects	50–50	1			
Total	1–50	**50**			

Net 35 or more: ready, move to Paper 3. **25–35:** re-read the weak topics and redo their quizzes. **Below 25:** 3–4 more days on Module 2, then retake after a week.